

MAT1575, Classwork8, Fall2026

3.2 Trigonometric Integrals (p.241-249)

1. Case1: Integrating Products and Powers of $\sin x$ and $\cos x$: $\int \sin^j(x) \cos(x) dx$ or $\int \cos^j(x) \sin(x) dx$

u -sub
Evaluate $\int \cos^3(x) \sin(x) dx$ $\xrightarrow{u=\cos(x), du=-\sin(x)dx}$ $\int u^3 (-du) = -\int u^3 du = -\frac{u^4}{4} + C$
 $= -\frac{\cos^4(x)}{4} + C$

u -sub
2. Case2: Integrating Products and Powers of $\sin x$ and $\cos x$: $\int \cos^j(x) \sin^k(x) dx$ where k is odd.

Evaluate $\int \cos^2(x) \sin^3(x) dx = \int \cos^2(x) \sin^2(x) \sin(x) dx = \int \cos^2(x) (1 - \cos^2(x)) \sin(x) dx$
 $u = \cos(x), du = -\sin(x)dx$
 $= \int u^2 (1 - u^2) (-du) = -\int u^2 (1 - u^2) du$
 $= -\int u^2 - u^4 du = -\left[\frac{u^3}{3} - \frac{u^5}{5}\right] + C = -\frac{\cos^3(x)}{3} + \frac{\cos^5(x)}{5} + C$
 $(= \int u^4 - u^2 du)$

3. Case3: Integrating an Even Powers of $\sin x$ or $\cos x$: $\int \sin^k x dx$ or $\int \cos^k x dx$ where k is even. $\sin^2(x) = \frac{1 - \cos(2x)}{2}$

Evaluate $\int \sin^2(x) dx = \int \frac{1 - \cos(2x)}{2} dx = \int \frac{1}{2} - \frac{\cos(2x)}{2} dx$
 $= \frac{x}{2} - \left(-\frac{\sin(2x)}{4}\right) + C$

4. Problem-Solving Strategy: Integrating Products and Powers of $\sin x$ and $\cos x$

To integrate $\int \cos^j(x) \sin^k(x) dx$, we have 3 cases:

- (a) k is odd: rewrite $\sin^k(x) = \sin^{k-1}(x) \cdot \sin(x)$ and use $\sin^2(x) = 1 - \cos^2(x)$ and $u = \cos(x)$
(b) j is odd: rewrite $\cos^j(x) = \cos^{j-1}(x) \cdot \cos(x)$ and use $\cos^2(x) = 1 - \sin^2(x)$ and $u = \sin(x)$
(c) both k and j are even: use $\sin^2(x) = \frac{1 - \cos(2x)}{2}$ and $\cos^2(x) = \frac{1 + \cos(2x)}{2}$, then go back to (a) or (b)

5. Case4: Integrating Products of Sines and Cosines of Different Angles $\sin(ax)$, $\sin(bx)$, $\cos(ax)$, $\cos(bx)$.

Evaluate $\int \sin(5x) \cos(3x) dx$. Hint: use sum of angle formula $\sin(ax) \cos(bx) = \frac{1}{2} (\sin((a-b)x) + \sin((a+b)x))$
 $= \int \frac{\sin(8x) + \sin(2x)}{2} dx = \int \frac{\sin(8x)}{2} + \frac{\sin(2x)}{2} dx$
 $= -\frac{\cos(8x)}{8 \cdot 2} - \frac{\cos(2x)}{2 \cdot 2} + C$

The other two: $\sin(ax) \sin(bx) = \frac{1}{2} (\cos((a-b)x) - \cos((a+b)x))$, and $\cos(ax) \cos(bx) = \frac{1}{2} (\cos((a-b)x) + \cos((a+b)x))$

6. Evaluate $\int \cos^8(x) \sin^5(x) dx$ $\xrightarrow{u=\cos(x), du=-\sin(x)dx}$ $= \int \cos^8(x) \sin^4(x) \sin(x) dx$
 $= \int \cos^8(x) (1 - \cos^2(x)) (1 - \cos^2(x)) \sin(x) dx$
 $= \int u^8 (1 - u^2) (1 - u^2) (-du)$
 $= \int -u^8 (1 - 2u^2 + u^4) du = \int -u^8 + 2u^{10} - u^{12} dx$
 $= -\frac{u^9}{9} + \frac{2u^{11}}{11} - \frac{u^{13}}{13} + C$
 $= -\frac{\cos^9(x)}{9} + \frac{2\cos^{11}(x)}{11} - \frac{\cos^{13}(x)}{13} + C$

7. Review: The known integrals involving $\tan(x)$ and $\sec(x)$.

$\int \sec^2(x) dx =$	$\int \sec(x) \tan(x) dx =$	$\int \tan(x) dx =$	$\int \sec(x) dx =$
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8. Evaluate $\int \sec^5(x) \tan(x) dx$ by using 7. directly.

9. Case1: Integrating $\int \tan^k(x) \sec^j(x) dx$ **when j is even** using $\sec^2(x) =$ and let $u =$

Evaluate $\int \tan^6(x) \sec^4(x) dx$

10. Case2: Integrating $\int \tan^k(x) \sec^j(x) dx$ **when k is odd** using $\tan^2(x) =$ and let $u =$

Evaluate $\int \tan^5(x) \sec^3(x) dx$

11. Case3: Integrating $\int \tan^k(x)$ when **k is odd , and $k \geq 3$** using $\tan^2(x) =$ and let

Evaluate $\int \tan^3(x) dx$

12. Case4: Integrating $\int \sec^j(x) dx$ **when j is odd and $j \geq 3$** using integration by parts and $\tan^2(x) = \sec^2(x) - 1$

Evaluate $\int \sec^3(x) dx$

13. Problem-Solving Strategy: **Integrating Products and Powers of $\tan x$ and $\sec x$**

To integrate $\int \sec^j(x) \tan^k(x) dx$, we have 4 cases:

(a) j is positive even: _____

(b) k is odd and $j \geq 1$: _____

(c) $k \geq 3$ is odd and $j = 0$: _____

(d) k is even and j is odd: _____

14. Reduction Formulas: $\int \sec^n(x) dx =$
 $\int \tan^n(x) dx =$

(which is the same as 13 (c))